

E079

Calculus III for Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

On a humid day, a forklift made in 2019 pushes a crate with force $F = (3, 4, 0)$ N while the crate moves $d = (2, 1, 5)$ m. How much work does the force do?

- A. 10 J
- B. 15 J
- C. About 27.4 J
- D. $(6, 4, 0)$ J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For what value of k are $(2, -1, k)$ and $(1, 4, 2)$ perpendicular?

- A. $k = 1$
- B. $k = -1$
- C. $k = 0$, because perpendicular vectors must share a zero component.
- D. $k = -3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A solar-panel supplier based in Arizona models a panel edge as $u = (1, 2, 0)$ m and a second edge as $v = (4, 1, 0)$ m. The panels carry a 25-year warranty. What is the area of the parallelogram the edges span?

- A. -7 m^2
- B. 6 m^2
- C. 3.5 m^2
- D. 7 m^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

The temperature on a metal plate, cooled by a fan bought online, is $f(x, y) = x^2y + 3y^2$. The plate is 40 cm wide. What is $\partial f / \partial x$ at (1, 2)?

- A. 16
- B. 2
- C. 4
- D. 13

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A hiker carrying a 12 kg pack stands at (1, 1) on terrain with height $f(x, y) = x^2 + 3y^2$. The trail was mapped by a volunteer club. What is the maximum rate of climb available from that point?

- A. 8
- B. 4
- C. $\sqrt{40} \approx 6.32$
- D. 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

At point P, $\nabla f = (3, -4)$. In which unit direction does f decrease fastest?

- A. $(-3/5, 4/5)$
- B. $(3/5, -4/5)$
- C. $(4/5, 3/5)$
- D. $(-3, 4)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A cost model built in a spreadsheet by a summer intern is $f(x, y) = x^2 + y^2 - 4x + 6y$. Its only critical point is (2, -3). The company sells in 14 countries. How should the point be classified?

- A. The lowest cost under every possible model change.
- B. A saddle point.
- C. A local maximum.
- D. A local minimum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

At a critical point, $f_{xx} = 3$, $f_{yy} = 2$, and $f_{xy} = 4$. What kind of point is it?

- A. A local maximum, because D is negative.
- B. A saddle point.
- C. A local minimum, because $D = 6 + 16 = 22$.
- D. A local minimum, because $f_{xx} > 0$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A farm crew will fence a rectangular pen, using cedar stained dark brown, so that $x + y = 20$ m, where x and y are the side lengths. They want to maximize $f = xy$. Using Lagrange multipliers, what is the largest possible value of xy ?

- A. 20 m^2
- B. 100 m^2
- C. 10
- D. No maximum; xy can grow without limit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A drone flies through the field $F = \nabla\phi$, where $\phi = x^2y$, from $(0, 0)$ to $(1, 2)$. Its path winds through three loops over a lake before arriving. How much work does the field do on the drone?

- A. -2
- B. 0
- C. It depends on the length of the looping path.
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A field does 5 J of work on an object moving once around a closed loop. What can you conclude about the field?

- A. The measurement is wrong, because every field does zero work around a loop.
- B. It is conservative.
- C. It is not conservative.
- D. Its divergence must be zero.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An engineer runs a simulation that took 6 hours on the field $F = (x, y, z)$ over an aluminum cube with sides of length 2. What is the net outward flux through the cube's surface?

- A. 0, because flux through a closed surface always cancels.
- B. 24
- C. 3
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

~~On a humid day,~~ a forklift ~~made in 2019~~ pushes a crate with force $F = (3, 4, 0)$ N while the crate moves $d = (2, 1, 5)$ m. How much work does the force do?

Target word: work as a dot product **Decides:** $W = F \cdot d = 3(2) + 4(1) + 0(5)$.

A	Correct	$6 + 4 + 0 = 10$ J.
B	Wrong stem number	Adds the 5 m z-displacement as if the force had a z-component.
C	Dropped element	$ F d = 5 \times \sqrt{30}$ ignores the angle between the vectors.
D	Adjacent category	Multiplies components but never sums them; work is a scalar.

2. Answer A Concept 1 • transfer

For what value of k are $(2, -1, k)$ and $(1, 4, 2)$ perpendicular?

Target word: solve for k from orthogonality **Decides:** $2 - 4 + 2k = 0$.

A	Correct	$2k = 2$, so $k = 1$.
B	Inverted operation	Sign error when moving -2 across the equals sign.
C	Absolute language	Perpendicularity requires a zero dot product, not a zero component.
D	Dropped element	Drops the minus sign on -1 ($2 + 4 + 2k = 0$).

3. Answer D Concept 2

A solar-panel supplier ~~based in Arizona~~ models a panel edge as $u = (1, 2, 0)$ m and a second edge as $v = (4, 1, 0)$ m. ~~The panels carry a 25-year warranty.~~ What is the area of the parallelogram the edges span?

Target word: parallelogram area **Decides:** $u \times v = (0, 0, 1 \cdot 1 - 2 \cdot 4) = (0, 0, -7)$.

A	Inverted operation	Reports the signed component; area is the magnitude.
B	Confusion set	$u \cdot v = 4 + 2 = 6$ is the dot product, not an area.
C	Adjacent category	Half the cross-product length is the triangle, not the parallelogram.
D	Correct	$ u \times v = -7 = 7$.

4. Answer C Concept 3

The temperature on a metal plate, ~~cooled by a fan bought online,~~ is $f(x, y) = x^2y + 3y^2$. ~~The plate is 40 cm wide.~~ What is $\partial f / \partial x$ at $(1, 2)$?

Target word: partial derivative in x **Decides:** $\partial f / \partial x = 2xy$; the $3y^2$ term is constant in x .

A	Adjacent category	Also differentiates $3y^2$ as if it varied with x ($2xy + 6y$).
B	Dropped element	$2x$ drops the y factor.
C	Correct	$2(1)(2) = 4$.
D	Confusion set	$\partial f / \partial y = x^2 + 6y = 1 + 12 = 13$.

5. Answer C Concept 4

A hiker carrying a 12 kg pack stands at (1, 1) on terrain with height $f(x, y) = x^2 + 3y^2$. The trail was mapped by a volunteer club. What is the maximum rate of climb available from that point?

Target word: maximum directional derivative **Decides:** $\nabla f(1, 1) = (2, 6)$; maximum rate = $|\nabla f|$.

A	Adjacent category	Adds components instead of taking the length.
B	Confusion set	The height $f(1, 1)$, not a rate.
C	Correct	$\sqrt{(4 + 36)} = \sqrt{40}$.
D	Dropped element	Forgets the square root.

6. Answer A Concept 4 • transfer

At point P, $\nabla f = (3, -4)$. In which unit direction does f decrease fastest?

Target word: steepest descent direction **Decides:** Descent is along $-\nabla f$, normalized by $|\nabla f| = 5$.

A	Correct	$-\nabla f / \nabla f = (-3, 4) / 5$.
B	Inverted operation	That is the direction of fastest increase.
C	Confusion set	Perpendicular to ∇f , where the rate of change is zero.
D	Dropped element	Right direction but not a unit vector.

7. Answer D Concept 5

A cost model built in a spreadsheet by a summer intern is $f(x, y) = x^2 + y^2 - 4x + 6y$. Its only critical point is (2, -3). The company sells in 14 countries. How should the point be classified?

Target word: classify the critical point **Decides:** $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 4 > 0$ and $f_{xx} > 0$.

A	Absolute language	The test classifies this function's point; it does not guarantee results for other models.
B	Confusion set	Saddles need $D < 0$.
C	Inverted operation	Reverses the sign rule on f_{xx} .
D	Correct	$D > 0$ with $f_{xx} > 0$.

8. Answer B Concept 5 • transfer

At a critical point, $f_{xx} = 3$, $f_{yy} = 2$, and $f_{xy} = 4$. What kind of point is it?

Target word: classify from given second partials **Decides:** $D = 3(2) - 4^2 = -10 < 0$.

A	Inverted operation	A negative D rules out both extrema.
B	Correct	$D < 0$ always means a saddle.
C	Wrong stem number	Adds $(f_{xy})^2$ instead of subtracting it.
D	Dropped element	Checks f_{xx} without first checking D .

9. Answer B Concept 6

A farm crew will fence a rectangular pen, ~~using cedar stained dark brown~~, so that $x + y = 20$ m, where x and y are the side lengths. They want to maximize $f = xy$. Using Lagrange multipliers, what is the largest possible value of xy ?

Target word: constrained maximum of f **Decides:** $\nabla f = \lambda \nabla g$ gives $y = \lambda$ and $x = \lambda$, so $x = y = 10$.

A	Wrong stem number	Reports the constraint total, not the product.
B	Correct	$10 \times 10 = 100$.
C	Confusion set	That is λ , not the maximum of f .
D	Dropped element	Ignores the constraint $x + y = 20$.

10. Answer D Concept 7

A drone flies through the field $F = \nabla \phi$, where $\phi = x^2y$, from $(0, 0)$ to $(1, 2)$. ~~Its path winds through three loops over a lake before arriving.~~ How much work does the field do on the drone?

Target word: work in a conservative field **Decides:** Work = $\phi(1, 2) - \phi(0, 0)$.

A	Inverted operation	Computes $\phi(\text{start}) - \phi(\text{end})$.
B	Confusion set	Zero applies to closed loops; this path ends at a different point.
C	Camouflage	Conservative-field work is path independent.
D	Correct	$1^2 \times 2 - 0 = 2$, regardless of path.

11. Answer C Concept 7 · transfer

A field does 5 J of work on an object moving once around a closed loop. What can you conclude about the field?

Target word: reverse: infer field type from loop work **Decides:** Conservative fields give zero work on every closed loop.

A	Absolute language	Only conservative fields do.
B	Inverted operation	Reverses the closed-loop test.
C	Correct	Nonzero work on a closed loop rules out a potential function.
D	Adjacent category	Divergence concerns flux, not circulation.

12. Answer B Concept 8

An engineer runs a simulation ~~that took 6 hours~~ on the field $F = (x, y, z)$ over an ~~aluminum~~ cube with sides of length 2. What is the net outward flux through the cube's surface?

Target word: flux via divergence theorem **Decides:** $\text{div } F = 3$; volume = $2^3 = 8$.

A	Absolute language	Net flux is zero only when the enclosed divergence totals zero.
B	Correct	$3 \times 8 = 24$.
C	Adjacent category	Reports the divergence, a pointwise density, not the total flux.
D	Dropped element	Uses the volume but drops $\text{div } F = 3$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Dot product, work, and orthogonality	$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$ is a scalar; $\text{work} = F \cdot d$; $a \cdot b = 0$ means the vectors are perpendicular.	Returning a vector of component products, or multiplying magnitudes and ignoring the angle.	Force and displacement vectors, or “perpendicular.”
2	Cross product and area	$a \times b$ is perpendicular to both; $ a \times b $ is the area of the parallelogram they span (half of it for the triangle).	Using the dot product for area, or reporting a signed component as an area.	Area spanned by two vectors, or a normal vector to a plane.
3	Partial derivatives	$\partial f / \partial x$ treats every other variable as a constant.	Differentiating terms that contain only the other variable.	A multivariable function and a rate “with respect to x.”
4	Gradient and directional derivative	∇f points in the direction of steepest increase, its length is the maximum rate, and $D_u f = \nabla f \cdot u$ for a unit vector u .	Adding gradient components for the maximum rate, or using a non-unit direction.	“Fastest increase,” “steepest,” or a rate in a given direction.
5	Second-derivative test	$D = f_{xx}f_{yy} - (f_{xy})^2$: $D > 0$ with $f_{xx} > 0$ is a local min, $D > 0$ with $f_{xx} < 0$ a local max, $D < 0$ a saddle.	Calling every critical point an extremum.	A critical point and a request to classify it.
6	Lagrange multipliers	At a constrained extremum $\nabla f = \lambda \nabla g$; solve with the constraint, then evaluate f .	Reporting λ as the optimum, or optimizing without the constraint.	“Subject to” a fixed budget, perimeter, or total.
7	Line integrals in conservative fields	If $F = \nabla \phi$, work from A to B equals $\phi(B) - \phi(A)$, independent of path; around any closed loop it is 0.	Thinking a longer or loopier path changes the work in a conservative field.	A potential function or a path that winds, loops, or returns to its start.
8	Divergence theorem	Net outward flux through a closed surface equals the triple integral of $\text{div } F$ over the enclosed volume.	Assuming net flux through a closed surface is always zero.	Flux out of a closed box, sphere, or cylinder.

Course scope (public description): This course extends calculus into multiple dimensions and vector analysis, emphasizing mathematical tools essential for engineering applications. Learners apply vector operations in two and three dimensions and use multivariable differentiation techniques to model and solve optimization and engineering problems. The course also focuses on analyzing vector fields through line, surface, and volume integrals to model physical and computational systems. Applications include engineering systems, physics based modeling, and optimization problems relevant to artificial intelligence and related engineering fields.